Semiconductors and How a Transistor Works

Introduction to Semiconductors

A semiconductor is a material whose electrical conductivity lies between that of a conductor, such as copper, and an insulator, such as glass or rubber. Its most important property is that its conductivity can be controlled precisely by adding tiny amounts of impurities, by applying voltage, or by changing temperature or light. This controllability makes semiconductors the foundation of all modern electronics, including smartphones, computers, televisions, cars, satellites and medical equipment. The most common semiconductor materials are silicon (Si) and germanium (Ge); compound semiconductors such as gallium arsenide (GaAs), silicon carbide (SiC) and gallium nitride (GaN) are used in high-frequency and high-power devices.

Why Is Silicon Widely Used?

Silicon is the most widely used semiconductor for several reasons. It is extremely abundant: silicon is the second most common element in the Earth's crust, found in sand and quartz as silicon dioxide. It can be purified to very high levels, forming single crystals. It forms a stable, high-quality insulating layer of silicon dioxide (SiO2) naturally when heated in oxygen, which is essential for building transistors. It works at higher temperatures than germanium, since its band gap of about 1.1 electron-volts keeps leakage current low. It is also cheap and well understood after decades of manufacturing experience. For these reasons, the region in California where many chip companies grew is known as Silicon Valley.

Intrinsic and Extrinsic Semiconductors: Doping

Pure silicon is called an intrinsic semiconductor and conducts poorly at room temperature. Its conductivity is greatly increased by doping, the deliberate addition of small amounts of impurity atoms. In n-type silicon, atoms with five valence electrons, such as phosphorus or arsenic, are added; they provide extra free electrons, the majority charge carriers. In p-type silicon, atoms with three valence electrons, such as boron, are added; they create "holes", places where an electron is missing, which act as positive charge carriers. When p-type and n-type regions are joined, a p-n junction forms. A p-n junction allows current to flow easily in one direction (forward bias) but blocks it in the other (reverse bias). This is the principle of the diode, and it is also the basis of LEDs and solar cells.

History of the Transistor

The transistor was invented in December 1947 at Bell Laboratories in New Jersey by John Bardeen, Walter Brattain and William Shockley. They received the Nobel Prize in Physics in 1956. Transistors replaced bulky, hot and fragile vacuum tubes, allowing electronics to become small, cheap and reliable. In 1958 Jack Kilby at Texas Instruments built the first integrated circuit, and in 1959 Robert Noyce at Fairchild Semiconductor developed a practical silicon version. In 1965 Gordon Moore observed that the number of transistors on a chip doubles roughly every two years, a trend called Moore's Law. Modern chips contain tens of billions of transistors, each only a few nanometres in critical dimensions.

How a Transistor Works

A transistor is a semiconductor device used as an electronic switch or as an amplifier. As a switch, a small signal turns a larger current on or off, representing the 1s and 0s of digital logic. As an amplifier, a small input signal controls a larger output signal.

The Bipolar Junction Transistor (BJT) is built from three layers of doped semiconductor, either NPN or PNP. The three terminals of a BJT are the emitter, the base and the collector. The emitter supplies charge carriers, the thin base controls the flow, and the collector gathers the carriers. In an NPN transistor, a small current into the base allows a much larger current to flow from the collector to the emitter. The BJT is therefore a current-controlled device, and the ratio of collector current to base current is called the current gain, or beta.

The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is the most common transistor in computer chips. It has three terminals called the gate, the source and the drain. A voltage applied to the gate creates an electric field through a thin oxide layer and forms a conducting channel between source and drain. The MOSFET is therefore voltage-controlled and uses very little power. CMOS technology combines N-type and P-type MOSFETs and is used in processors and memory. Newer designs such as FinFET and gate-all-around transistors help continue miniaturisation.

How Chips Are Made

Chip manufacturing starts with sand purified into ultra-pure silicon, grown into a large crystal ingot and sliced into thin discs called wafers. Photolithography uses light, and in advanced factories extreme ultraviolet (EUV) light, to print circuit patterns onto the wafer. Steps of deposition, etching and doping are repeated many times over weeks. A modern factory called a fab costs billions of dollars. Taiwan's TSMC is the world's largest contract chipmaker, and the Dutch company ASML is the only maker of EUV lithography machines.

Semiconductors in India

India designs a large share of the world's chips; many global companies run chip design centres in Bengaluru, Hyderabad, Chennai and Noida. In December 2021 the government launched the India Semiconductor Mission with an outlay of Rs. 76,000 crore. Approved projects include a fabrication plant by Tata Electronics with Taiwan's PSMC at Dholera in Gujarat, and assembly and test facilities by Micron at Sanand in Gujarat and by Tata at Assam. The Semi-Conductor Laboratory (SCL) in Mohali is a government chip facility. IIT Madras developed SHAKTI, an open-source RISC-V processor family.

Challenges and Future

Challenges include the physical limits of shrinking transistors, heat, enormous manufacturing cost, geopolitical concentration of fabs and the global chip shortage of 2020 to 2022. The future includes chiplets, 3D stacking, wide-band-gap materials like SiC and GaN for electric vehicles, and AI accelerators.

Quick Facts

- A semiconductor has conductivity between a conductor and an insulator, controllable by doping.
- The three terminals of a BJT are the emitter, base and collector.
- Silicon is widely used because it is abundant, forms a stable oxide and works well at higher temperatures.
- The transistor was invented at Bell Labs in December 1947 by Bardeen, Brattain and Shockley.
- India Semiconductor Mission was launched in December 2021 with Rs. 76,000 crore.